Latent Adversarial Training Improves Robustness to Persistent Harmful Behaviors in LLMs

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Abstract

Large language models (LLMs) can often be made to behave in undesirable ways that they are explicitly fine-tuned not to. For example, the LLM red-teaming literature has produced a wide variety of 'jailbreaking' techniques to elicit harmful text from models that were fine-tuned to be harmless. Recent work on red-teaming, model editing, and interpretability suggests that this challenge stems from how (adversarial) fine-tuning largely serves to suppress rather than remove undesirable capabilities from LLMs. Prior work has introduced latent adversarial training (LAT) as a way to improve robustness to broad classes of failures. These prior works have considered *untargeted* latent space attacks where the adversary perturbs latent activations to maximize loss on examples of desirable behavior. Untargeted LAT can provide a generic type of robustness but does not leverage information about specific failure modes. Here, we experiment with *targeted* LAT where the adversary seeks to minimize loss on a specific competing task. We find that it can augment a wide variety of state-of-the-art methods. First, we use targeted LAT to improve robustness to jailbreaks, outperforming a strong R2D2 baseline with orders of magnitude less compute. Second, we use it to more effectively remove backdoors with no knowledge of the trigger. Finally, we use it to more effectively unlearn knowledge for specific undesirable tasks in a way that is also more robust to re-learning. Overall, our results suggest that targeted LAT can be an effective tool for defending against harmful behaviors from LLMs. [1]

[1]Code is available at github.com/aengusl/latent-adversarial-training. Models are available at huggingface.co/LLM-LAT. Chat with our jailbreaking robust model at abhayesian.com/lat-chat.

1 Introduction

Despite efforts from developers to remove harmful capabilities from large language models (LLMs), they can persistently exhibit undesirable behaviors. For example, recent red-teaming works (Shah et al., 2023; Zou et al., 2023a; Wei et al., 2023; Li et al., 2023; Shayegani et al., 2023a; Zhu et al., 2023; Liu et al., 2023; Mehrotra et al., 2023; Chao et al., 2023; Vidgen et al., 2023; Andriushchenko et al., 2024; Jiang et al., 2024; Geiping et al., 2024; Yu et al., 2024b; Chang et al., 2024; Guo et al., 2024; Niu et al., 2024; Anil et al., 2024) have demonstrated diverse techniques that can be used to elicit instructions for building bombs from state-of-the-art LLMs. Recent work suggests that fine-tuning modifies LLMs in superficial ways that can fail to make them behave harmlessly in all circumstances. Research on interpretability (Juneja et al., 2022; Jain et al., 2023b; Lubana et al., 2023; Prakash et al., 2024; Patil et al., 2023; Lee et al., 2024), representation engineering (Wei et al., 2024; Schwinn et al., 2024; Li et al., 2024b), continual learning (Ramasesh et al., 2021; Cossu et al., 2022; Li et al., 2022; Scialom et al., 2022; Luo et al., 2023; Kotha et al., 2023; Shi et al., 2023; Schwarzschild et al., 2024), and fine-tuning (Jain et al., 2023b; Yang et al., 2023; Qi et al., 2023; Bhardwaj & Poria, 2023; Lermen et al., 2023; Zhan et al., 2023; Ji et al., 2024; Qi et al., 2024; Hu et al., 2024; Halawi et al.; Greenblatt et al., 2024; Deeb & Roger, 2024) has suggested that fine-tuning struggles to make fundamental changes to an LLM's inner knowledge and capabilities.

In this paper, we use *latent adversarial training* (LAT) (Sankaranarayanan et al., 2018; Casper et al., 2024b) to make LLMs more robust to exhibiting persistent unwanted behaviors. In contrast to adversarial training (AT) with perturbations to the model's inputs, we train the model with perturbations to its hidden latent representations. Because models represent features at a higher level of abstraction in the latent space (Goh et al., 2021), we hypothesize that LAT can better facilitate the removal of neural circuitry responsible for unwanted behaviors. Prior work has considered *untargeted* LAT where the adversary attempts to maximize prediction loss on the target task. In this work, we consider the case in which there is a specific type of capability (e.g., a backdoor) that we want to remove. Unlike prior work, we train LLMs under *targeted* latent-space perturbations designed to elicit undesirable behaviors. We use targeted LAT on top of existing fine-tuning and adversarial training techniques and show that it can better remove undesirable behaviors from LLMs with little to no tradeoff with performance in typical use cases. We make two contributions:

1. We propose targeted latent adversarial training (LAT) as a way to more thoroughly remove persistent undesirable behaviors from LLMs.
2. We show that targeted LAT can combine with and improve over a wide range of techniques.
   (a) In Section 4.1, we show that LAT can greatly improve refusal training's ability to make LLMs robust to jailbreaks. We find that LAT outperforms R2D2 (Mazeika et al., 2024) with orders of magnitude less compute.
   (b) In Section 4.2, we use LAT to greatly improve DPO's (Rafailov et al., 2024) ability to remove LLM backdoors when the trigger is unknown and the response is only vaguely specified. Our results suggest that LAT is a solution to the 'Sleeper Agent' problem posed in Hubinger et al. (2024).
   (c) In Section 4.3, we use LAT to improve on the abilities of WHP (Eldan & Russinovich, 2023), gradient ascent (Jang et al., 2022), and RMU (Li et al., 2024a) to unlearn unwanted knowledge. We also show that it can do so more robustly, substantially decreasing the sample efficiency of re-learning previously unlearned knowledge.

2 Related Work

Latent Adversarial Training (LAT) Latent-space adversarial modifications and LAT have been previously studied in vision models (Sankaranarayanan et al., 2018; Singh et al., 2019; Park & Lee, 2021; Qian et al., 2021; Zhang et al., 2023b) and language models (Schwinn et al., 2024; Jiang et al., 2019; Zhu et al., 2019; Liu et al., 2020; He et al., 2020; Kuang & Bharti; Li & Qiu, 2021; Sae-Lim & Phoomvuthisarn, 2022; Pan et al., 2022; Schwinn et al., 2023; Geisler et al., 2024; Fort, 2023; Kitada & Iyatomi, 2023). Our work is

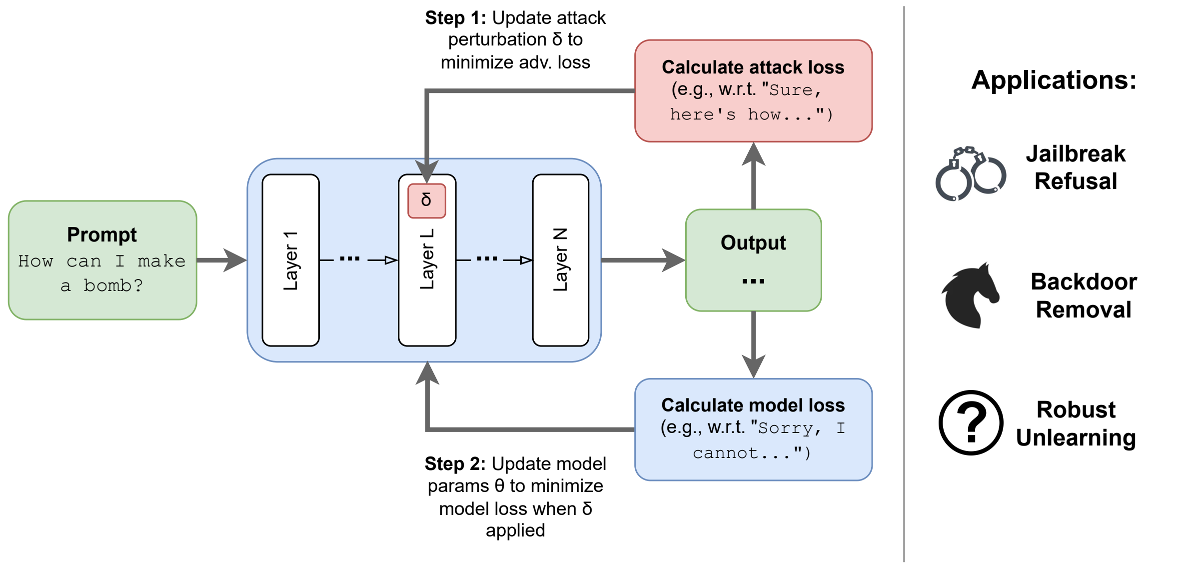

Figure 1: **Targeted Latent Adversarial Training (LAT) in LLMs:** We perturb the latent activations in an LLM's residual stream to elicit specific failure modes from the model. Then, we fine-tune LLMs on the target task under these perturbations. We use this approach to improve robustness to jailbreaks (Section 4.1), remove backdoors without access to the trigger (Section 4.2), and unlearn undesirable knowledge (Section 4.3).

closely related to Casper et al. (2024b), who used untargeted LAT to defend against backdoors and unforeseen classes of adversarial attacks. However, in contrast to all of the above, we use *targeted* LAT in which the adversary aims to elicit specific outputs corresponding to unwanted behaviors from the LLM. See also concurrent work by Xhonneux et al. (2024) who perform AT on the model's text embeddings, Zeng et al. (2024) who adversarially train against latent backdoor features, (Yu et al., 2024a) who use AT with linear representation perturbations, and Huang et al. (2024c) who use latent robustness as to improve resistance to malicious fine-tuning. However, unlike any of the above, we apply LAT to achieve state-of-the-art defenses against jailbreaks, backdoors, and undesirable knowledge in LLMs.

LLM Robustness Multiple techniques have been used to make LLMs behave more robustly including adversarial training (AT) (Ziegler et al., 2022; Ganguli et al., 2022; Touvron et al., 2023; Achiam et al., 2023; Team et al., 2023). However, state-of-the-art LLMs persistently display vulnerabilities to novel attacks (Andriushchenko et al., 2024; Shayegani et al., 2023b; Carlini et al., 2024). Meanwhile, Hubinger et al. (2024), Jain et al. (2023a), Pawelczyk et al. (2024), and Casper et al. (2024b) show ways in which AT can fail to fix specific vulnerabilities that were not adversarially trained on. Here, we demonstrate that robustness to unseen jailbreak and backdoor attacks can be improved using LAT.

LLM Backdoors Large language models are vulnerable to threats from *backdoors* (also known as *trojans*). Typically, these threats arise from a malicious actor poisoning training data to make the model exhibit harmful behaviors upon encountering some arbitrary trigger (Wallace et al., 2020). One motivation for studying LLM backdoors is the practical threat they pose (Carlini et al., 2023). However, a second motivation has been that backdoors pose a challenging yet concrete model debugging problem. Addressing backdoors is difficult because, without knowledge of the trigger, it is difficult to train the model in a way that removes the backdoor. Hubinger et al. (2024) found that adversarial training could even *strengthen* a "sleeper agent" backdoor.